Interaction

COMP7507
Visualization & Visual Analytics



SCHOOL OF
COMPUTING &
DATA SCIENCE
The University of Hong Kong

Interaction & Distortion Techniques

- Interaction:
  - Allows the user to interact with the visualization and dynamically change the visualizations according to the exploration objectives (as compared to static visualization on paper)
  - Enables relating and combining multiple independent visualizations
- Distortion:
  - Allows focusing on details while preserving an overview of the data
  - Shows portions of the data with a high level of detail, while others are shown in a lower level of detail

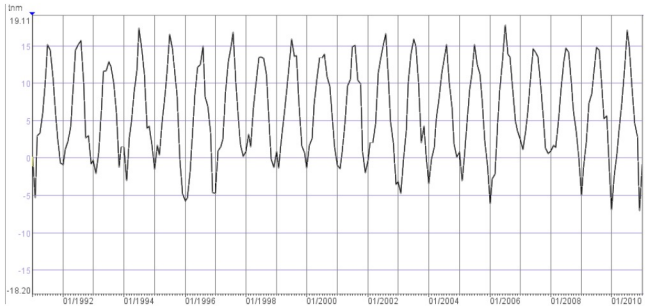
Use of Interaction

- **Interaction** is useful for integrating human in the data exploration process and applying its perceptual abilities
  - Changing data representation
  - Data transformation
  - Data selection and filtering
  - Relating multiple graphical views
  - Focusing and getting details

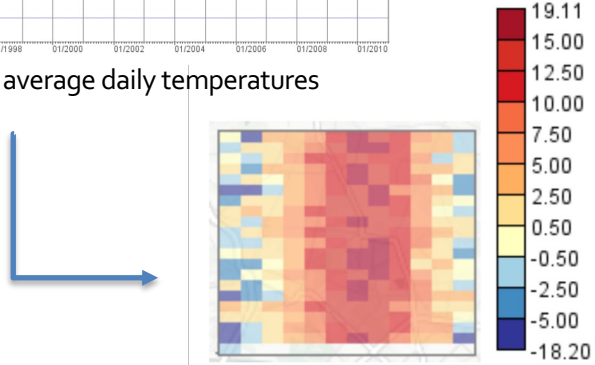
Interaction for

Changing Data Representation

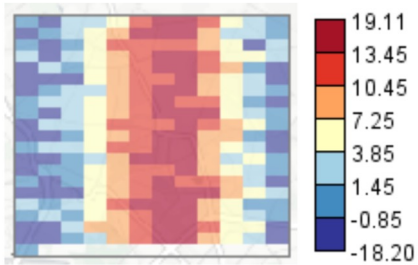
- needed when we want to look at the same data from different task-specific perspectives



monthly averages of average daily temperatures



rows for the years 1991-2011;
columns for the months (Jan-Dec)



different bins

[Andrienko et al.]

Interaction for

Data Transformation

- Different kinds of data transformation: discretization, logarithmic transformation, aggregation, merging, smoothing, etc.
- Various purposes: to get a clearer view, to simplify display or reduce amount of data, to disregard excessive details, etc.

Interaction for

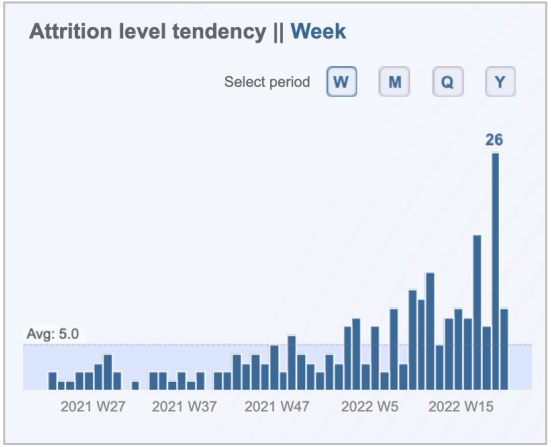
Data Selection and Filtering

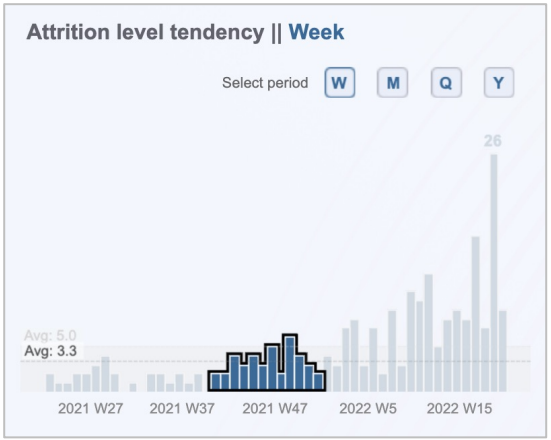
- Selection: to take a portion of data and visualize (or highlight) this portion only
- Filtering: to remove data that is currently out of interest from the visual views. (usually need data selection before filtering)
- Use interaction to specify the selection/filtering criteria
  - Attribute values, temporal/spatial positions, related to a particular entity or some other data
- Need UI elements (e.g., sliders)

Interaction for

Data Selection and Filtering

- It is important that we may highlight the data subset of interest while keeping the remaining data visible so that we can see how the interested data relates to the others.
- Highlighting can be done by brushing, which is typically done by dragging the mouse cursor over an area in the visualization.





Interaction for

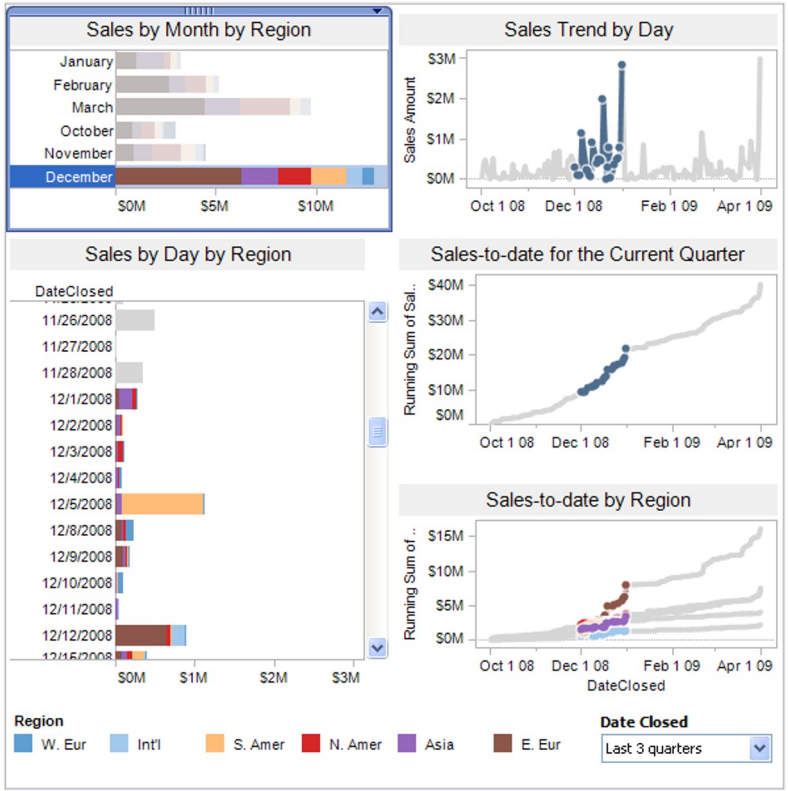
Relating Multiple Graphical Views

- Needed when we want to look at the same data from different perspectives at the same time
- Coordinated multiple views (CMV) – data displays are coordinated, easy to find corresponding pieces of information from different displays
- Facilitates visual analytics, e.g., detection of dependencies and correlations.
- Brushing and Filtering are some techniques that allow for coordination

Interaction for

Relating Multiple Graphical Views

- Linking & Brushing
  - Link data in different visualization views
  - Brushed points are highlighted in all visualizations of the same data set

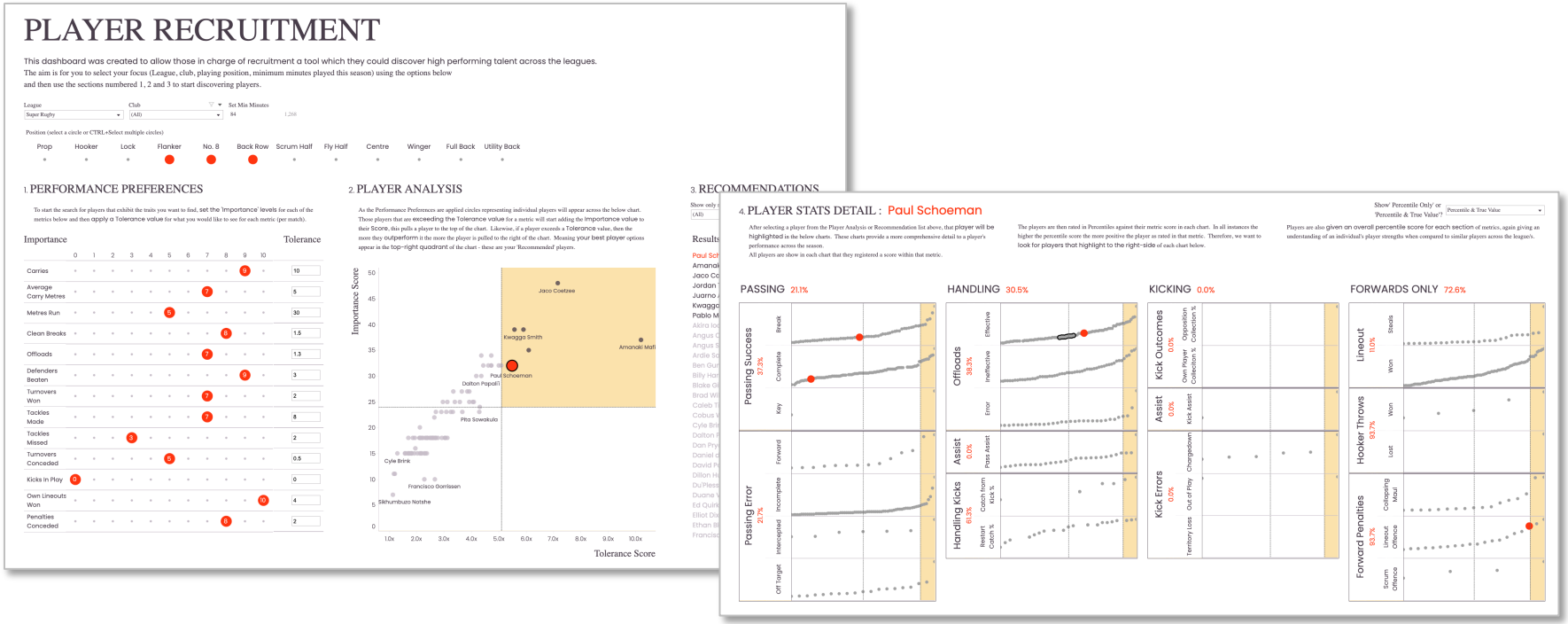


["Enhancing Visual Analysis by Linking Multiple Views of Data", Tableau whitepaper.]

Advanced Data Visualization : Adding Linking and Brushing Capability
https://www.youtube.com/watch?v=O1-w9pZeiTU

MyBrush: Brushing and Linking with Personal Agency (InfoVis 2017 paper)
https://www.youtube.com/watch?v=1iAtPhWEV9I

A Tableau Example



https://public.tableau.com/app/profile/sparsonsdataviz/viz/PlayerRecruitmentTool/PlayerDiscoveryTool

Interaction for

Focusing and Getting Details

- needed when we want to change focus of interest and to access additional information.
- Several techniques:
  - **overview+detail**: spatial separation
  - **zooming**: temporal separation
  - **focus+context**: seamless focus in context
- Interaction and distortion work together

Overview + Detail

- Simultaneous display of an overview and a detailed view of data, but each in distinct display space.



Detail

User interacts with the two views separately

Overview

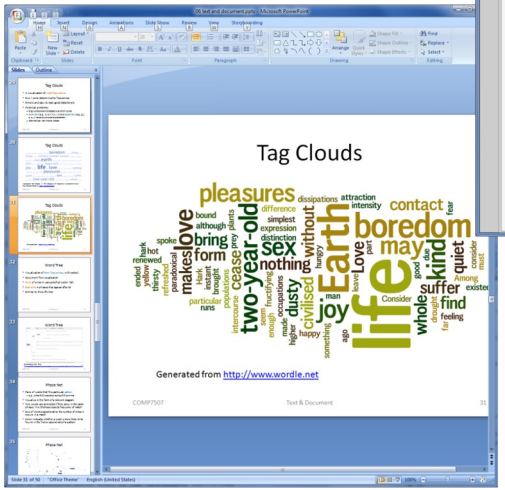
[Cockburn et al. 2002]

Asymmetric synchronization

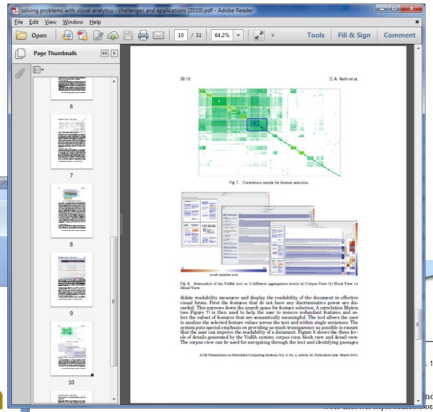
- Changes in the detailed view shows immediately in the overview
- Exploration in the overview does not change the detail view

Overview + Detail

- More examples

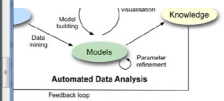


Microsoft PowerPoint

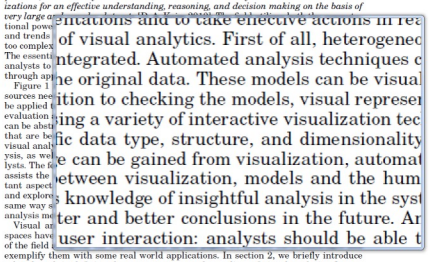


Adobe Acrobat

Magnifier on Windows

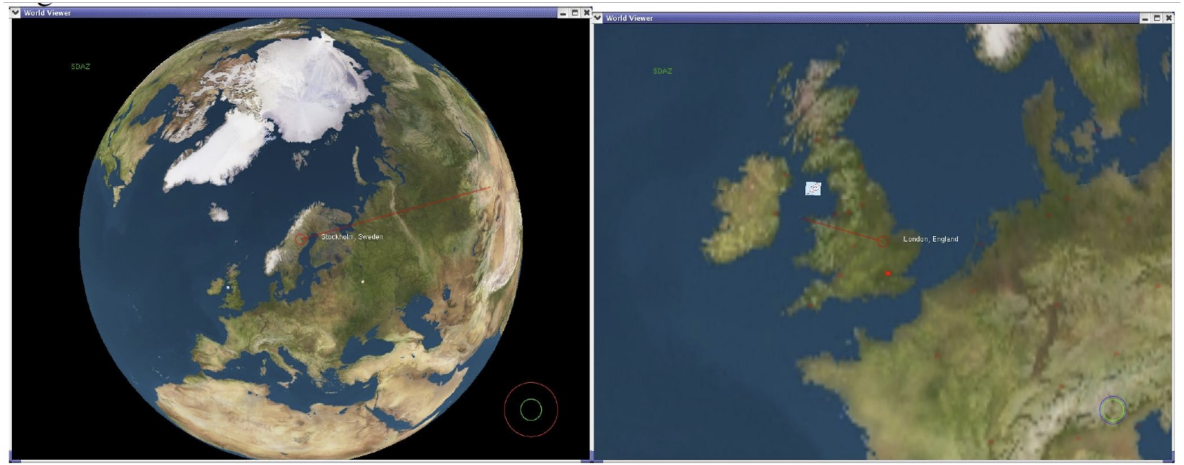


1. Visual Analytics Process [D. A. Keim 2010].



Zooming

- Use a single window to show focused and contextual views with temporal separation
- Magnify (zoom in to filter and thereby focus) and Demagnify (zoom out that acts as aggregation for context or for ag)
- Visual effect is similar to a geometric zooming done by moving a virtual camera to or farther from the objects.



[Cockburn et al. 2002]

Focus + Context

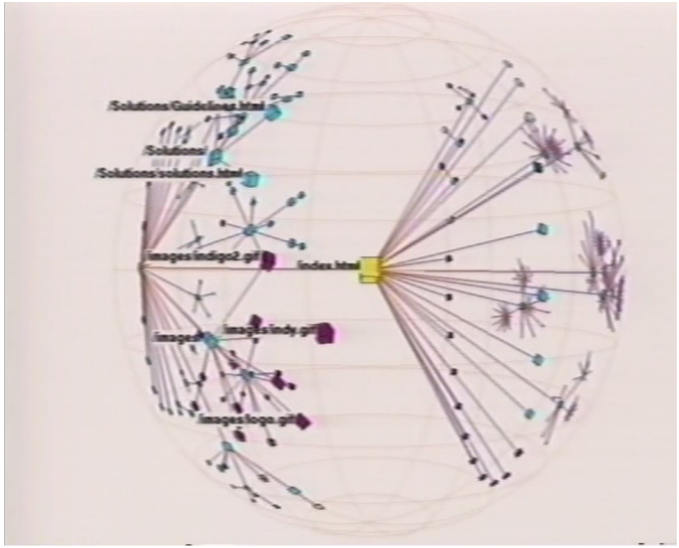
- Issues with O+D and Zooming: cognitive burden for users to correlate different views
- Focus + Context: focus within surrounding context in a single view
- Space can be stretched and squeezed, e.g., fisheye (distortion-based)
- Examples
  - Hyperbolic Tree Browser
  - Table Lens



[Cockburn et al. 2002]

Hyperbolic Tree Browser

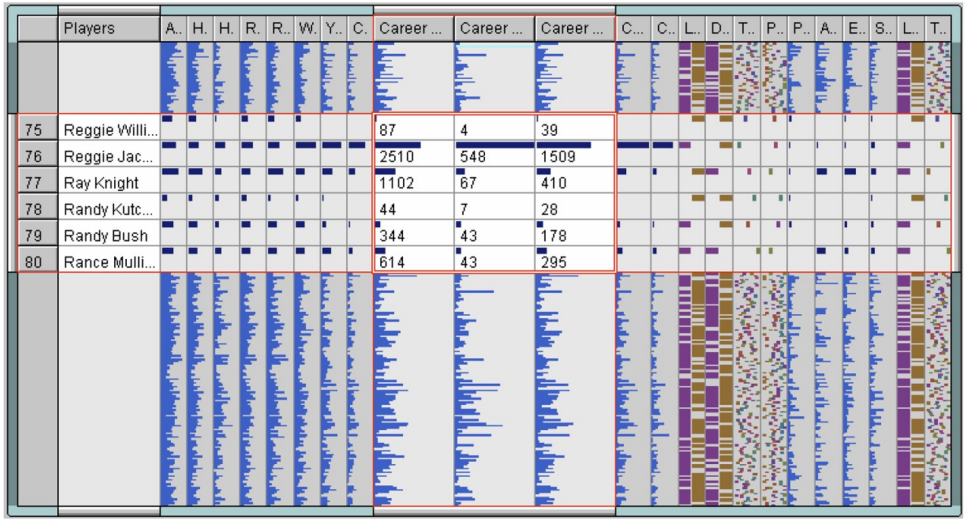
- Distortion based on hyperbolic geometry



[T. Munzner, "Drawing Large Graphs with H3Viewer and Site Manager," Graph Drawing'98.]

Table Lens

- Values of table entries are encoded as bars
- Fisheye: expand selected rows and columns
- Allow multiple focal points
- A kind of semantic zooming: the visual representation adapts to different scales (determined by the number of pixels available)



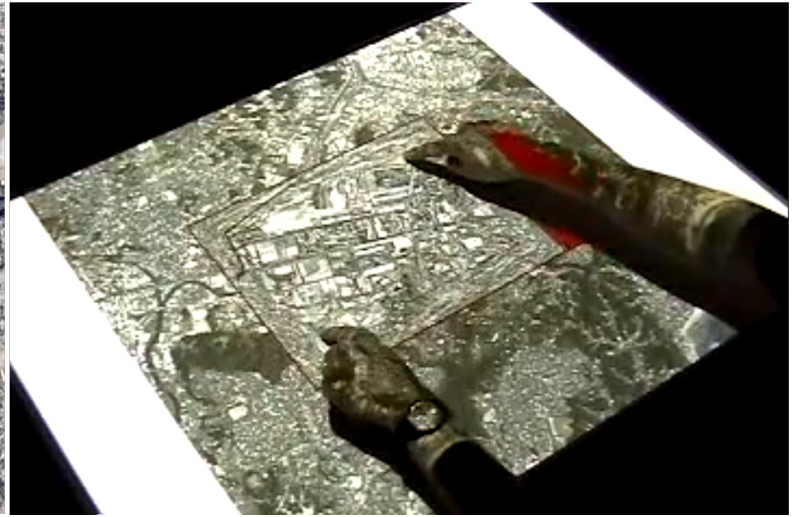
[Rao and Card, "The table lens: merging graphical and symbolic representations in an interactive focus + context visualization for tabular information," CHI'94.]

[http://www.youtube.com/watch?v=qWqTrRAC52U]

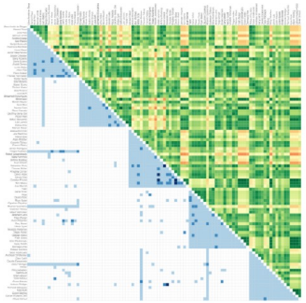
More Examples

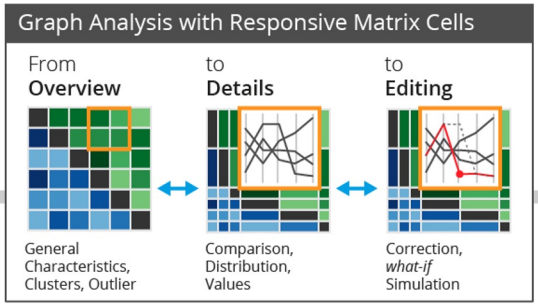


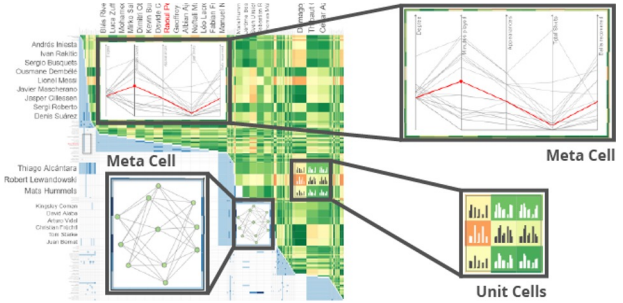
Rubber sheet map distortion
[http://www.youtube.com/watch?v=nTqlJdK3kcY]



Pliable display on a table
[http://www.youtube.com/watch?v=moLNHtRbaCA]







VIS 2020 paper:
Responsive Matrix Cells: A Focus+Context Approach for Multivariate Graphs
https://mt.inf.tu-dresden.de/en/research/research-projects/responsive-matrix-cells/
https://youtu.be/uOIB079aXAI?si=A3Df9zFqbNEMSB04&t=348

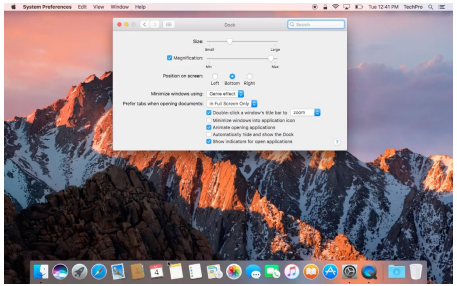
An example of semantic zooming

Problems of Distortion

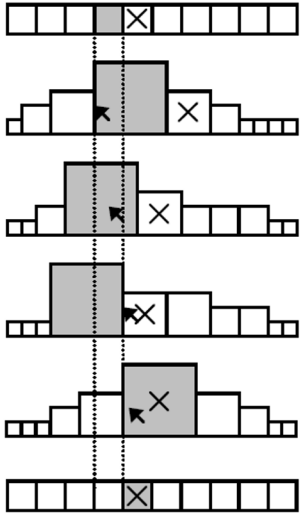
- Not suitable if spatial judgment is needed
- Difficult for target acquisition
  - Items are displayed away from screen location



(a) The Mac Os X Dock icon-panel.



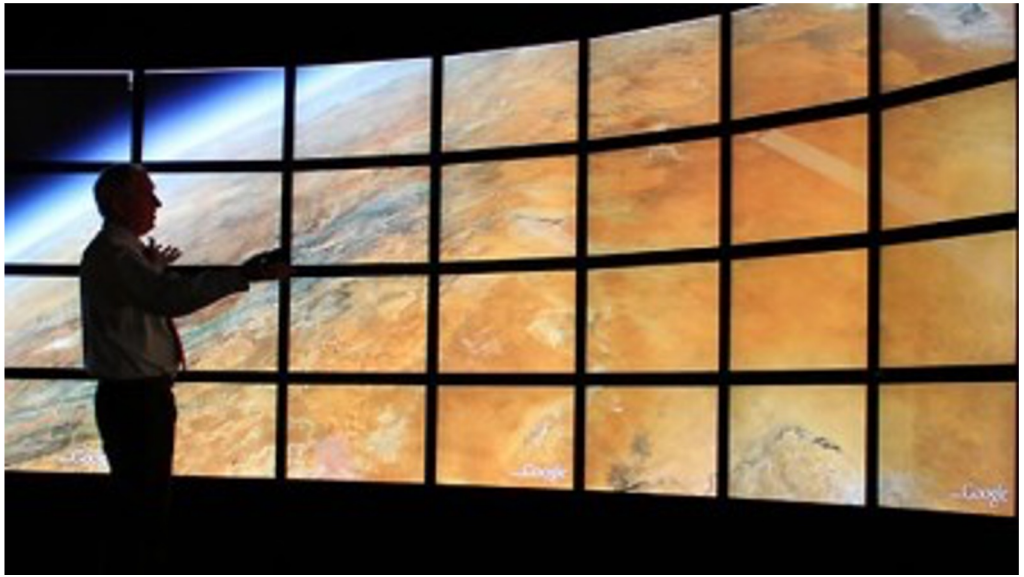
[Cockburn et al. 2002]



(b) Target movement caused by distortion. Items can be displayed at different locations (middle rows) from the actual locations needed to select them (top and bottom rows).

Focus+Context Without Distortion

- Wide field of view systems



Multiple large high-resolution displays



Mixed resolution displays

[Cockburn et al. 2002, Baudisch 2002]

References

- N. Andrienko, G. Andrienko, G. Fuchs, A. Slingsby, C. Turkay, S. Wrobel, "Visual Analytics for Data Scientists", 2020. (Chapter 3.3)
- A. Cockburn, A. Karlson, and B. B. Bederson, "A review of overview+detail, zooming, and focus+context interfaces," *ACM Computing Surveys*, 41(1), 2008.
- B. Shneiderman, "The Eye Have It: A Task by Data Type Taxonomy for Information Visualizations," *Visual Languages*, 1996.
- D. Keim, "Information Visualization and Visual Data Mining," *IEEE Transactions on Visualization and Computer Graphics*, 8(1), 2002, pp. 1 – 8.
- Colin Ware, *Information Visualization: Perception for Design*, 3rd ed., 2013. (Chapter 10)